Uncertainty and Model Checking for Atmospheric Inversions

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2026 BEACO2N Partner Meeting

What we know, and what we want to know

For the inversions we consider, we know:

$$\begin{aligned}
\mathbf{y} &: \text{Observed measurements} \quad && \text{dimension } M \\
\mathbf{x} &: \text{Unobserved flux} && \text{dimension } N
\end{aligned}$$

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\mathbf{H} &: \text{Footprints} \quad && \text{dimension } M \times N \\
\mathbf{B} &: \text{Prior covariance} && \text{dimension } N \times N \\
\mathbf{R} &: \text{Observation error} && \text{dimension } M \times M \\
\mathbf{m} &: \text{Prior inventory} && \text{dimension } N
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\mathbf{x} &\sim \mathcal{N}\left(\mathbf{m}, \mathbf{B}\right) && \text{Actual fluxes are smooth, centered at inventory} \\
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- Measurements of posterior uncertainty. (E.g., how large is $\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x})$?)
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Takeaway: All these are available from stuff you computed to get $\hat{\mathbf{x}}$.

Inference

Data generating process:

$$\mathbf{x} \sim \mathcal{N}\left(\mathbf{m}, \mathbf{B}\right) \qquad \text{Actual fluxes are smooth, centered at inventory}$$

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Inference: $\mathcal{P}(\mathbf{y}, \mathbf{x})$ is Gaussian $\Rightarrow \mathcal{P}(\mathbf{x}|\mathbf{y})$ is Gaussian.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\hat{\mathbf{x}}, \hat{\mathbf{V}}\right) \quad \text{where}$$

$$\hat{\mathbf{x}} = \mathbf{m} + \left(\mathbf{HB}\right)^{\intercal}\left(\mathbf{HBH}^{\intercal} + \mathbf{R}\right)^{-1}\left(\mathbf{y} - \mathbf{Hm}\right)$$

$$\hat{\mathbf{V}} = \mathbf{B} - \left(\mathbf{HB}\right)^{\intercal}\left(\mathbf{HBH}^{\intercal} + \mathbf{R}\right)^{-1}\left(\mathbf{HB}\right)$$

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Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}[\mathbf{x}]$. This is slow because of

- Forming $\mathbf{H}$ and $\mathbf{B}$ in memory
- Computing $\mathbf{HB}$ and $\mathbf{HBH}^\intercal$
- Inverting $(\mathbf{HBH}^\intercal + \mathbf{R})$

Sequoia has sped these up, and also saved to disk what she can.

Assume we can re-use these computations. Let's get posterior uncertainty, prior sensitivity, and data influence.

Posterior uncertainty

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\hat{\mathbf{x}}, \hat{\mathbf{V}}\right) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Problem: Obviously we can't compute $\hat{\mathbf{V}}$, it is huge.

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Example: Want s, the total emissions in hour t, summed over all spatial locations.

Let $\mathbf{a}$ be an $N-$vector with:

$$\mathbf{a}_n = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$s = \sum_{n=1}^{N} \mathbf{a}_n \mathbf{x}_n = \mathbf{a}^\intercal \mathbf{x} \Rightarrow$$

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(s\right) = \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(\mathbf{a}^\intercal \mathbf{x}\right) = \mathbf{a}^\intercal \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(\mathbf{x}\right) \mathbf{a}$$

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The covariance of the scalar s is a linear combination of stuff we computed for $\hat{\mathbf{x}}$:

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(s\right) = \mathbf{a}^\intercal \underbrace{\mathbf{B}}_{\text{done}} \mathbf{a} - (\underbrace{\mathbf{HB}}_{\text{done}} \mathbf{a})^\intercal \underbrace{(\mathbf{HBH}^\intercal + \mathbf{R})^{-1}}_{\text{done}} (\underbrace{\mathbf{HB}}_{\text{done}} \mathbf{a})$$

Variance (Tentative preliminary results!)

Posterior Variance Results

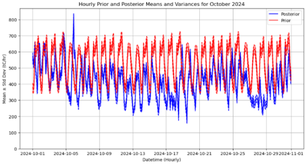

Figure 11: Hourly posterior mean and variance for a month of inversions

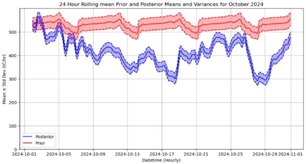

Figure 12: Smoothed hourly posterior mean and variance for a month of inversions

Prior sensitivity

Let ρ denote an assumed lengthscale for the prior.

The estimated $\hat{\mathbf{x}}$ depends on the prior $\mathbf{B}$, and the prior $\mathbf{B}$ depends on ρ.

Write $\hat{\mathbf{x}}(\rho)$ to denote this dependence.

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Idea: Starting from a solution at the initial guess ρ_0, form a Taylor series expansion

$$\hat{\mathbf{x}}(\rho) \approx \hat{\mathbf{x}} + \underbrace{\frac{\partial \hat{\mathbf{x}}(\rho)}{\partial \rho}}_{\substack{\textbf{The derivative is given easily} \\ \textbf{from stuff we already computed.}}} (\rho - \rho_0)$$

Prior sensitivity (Tentative preliminary results!)

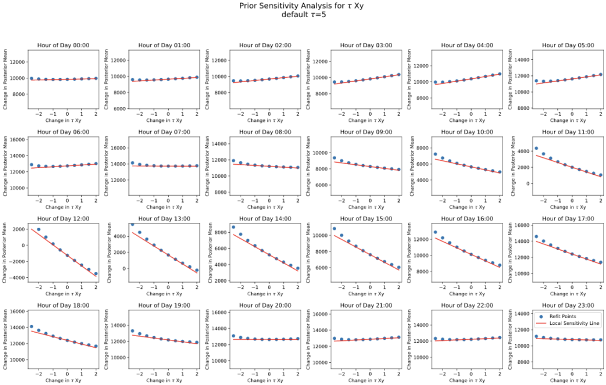

Figure 20: Prior sensitivity with respect to $\boldsymbol{\tau}_{\mathbf{xy}}$ for a single inversion

Data influence

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Let $\mathbf{r}$ be an N-vector encoding road pixels.

$$\mathbf{r}_n = \begin{cases} 1 & \text{If observation } n \text{ was a road pixel} \\ 0 & \text{otherwise} \end{cases}$$

Suppose we learn $\hat{\mathbf{x}}'$ with “roadier” observations $\mathbf{y}' = \mathbf{y} + \epsilon \mathbf{H}\mathbf{r}$.
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The inferred road pixel change is

$$\mathbf{r}^\intercal (\hat{\mathbf{x}}' - \hat{\mathbf{x}}) = \epsilon \mathbf{r}^\intercal \underbrace{(\mathbf{H}\mathbf{B})^\intercal \left(\mathbf{H}\mathbf{B}\mathbf{H}^\intercal + \mathbf{R}\right)^{-1} \mathbf{H}}_{\substack{\textbf{The gain matrix is given easily} \\ \textbf{from stuff we already computed.}}} \mathbf{r}$$

Like the covariance, we don't care about the whole gain matrix. We only really care about linear summaries, and these are easy to compute.

Data influence (Tentative preliminary results!)

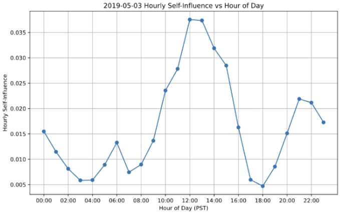

Figure 16: Hourly self-influence for a single inversion

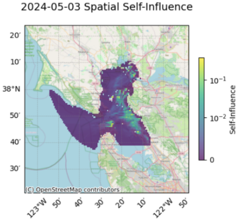

Figure 17: Spatial self-influence for a single inversion

Future work

By re-using quantities already computed for $\hat{\mathbf{x}}$, we can compute:

- Covariances of linear functions
- Derivatives with respect to model parameters (e.g. prior length scale)
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Straightforward mathematics, with lots of possibilities!